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GLM-5.2-FP8

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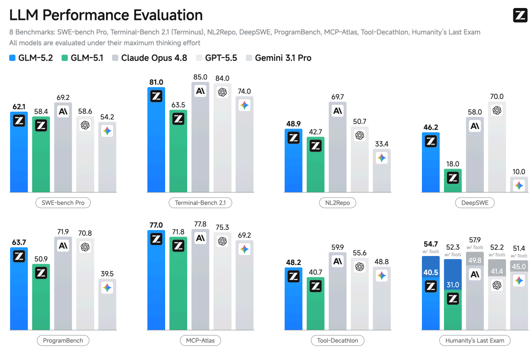
[Paper] [GitHub]

Introduction

We're introducing GLM-5.2, our latest flagship model for long-horizon tasks. It marks a substantial leap in long-horizon task capability over its predecessor GLM-5.1 and, for the first time, delivers that capability on a **solid 1M-token context**. GLM-5.2's new capabilities include:

- **Solid 1M Context:** A solid 1M-token context that stably sustains long-horizon work
- **Advanced Coding with Flexible Effort**: Stronger coding capabilities with multiple thinking effort levels to balance performance and latency
- **Improved Architecture**: We propose IndexShare, which reuses the same indexer across every four sparse attention layers, reducing per-token FLOPs by 2.9× at a 1M context length. We also improve GLM-5.2's MTP layer for speculative decoding, increasing the acceptance length by up to 20%

- **Pure Open**: An MIT open-source license — no regional limits, technical access without borders



Benchmark

Benchmark	GLM-5.2	GLM-5.1	Qwen3.7-Max	MiniMax M3
Reasoning				
HLE	40.5	31	41.4	37
HLE (w/ Tools)	54.7	52.3	53.5	-
CritPt	20.9	4.6	13.4	3.7
AIME 2026	99.2	95.3	97	-
HMMT Nov. 2025	94.4	94	95	84.4
HMMT Feb. 2026	92.5	82.6	97.1	84.4
IMOAnswerBench	91.0	83.8	90	-
GPQA-Diamond	91.2	86.2	90	93
Coding				
SWE-bench Pro	62.1	58.4	60.6	59
NL2Repo	48.9	42.7	47.2	42.1
DeepSWE	46.2	18	18	20
ProgramBench	63.7	50.9	-	-
Terminal Bench 2.1 (Terminus-2)	81.0	63.5	75	65
Terminal Bench 2.1 (Best Reported Harness)	82.7	69	-	-
FrontierSWE (Dominance)	74.4	30.5	-	-

Papers for `zai-org/GLM-5.2-FP8`

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Benchmark	GLM-5.2	GLM-5.1	Qwen3.7-Max	MiniMax M3
PostTrainBench	34.3	20.1	-	-
SWE-Marathon	13.0	1.0	-	-
Agentic				
MCP-Atlas (Public Set)	76.8	71.8	76.4	74.2
Tool-Decathlon	48.2	40.7	-	-

Serve GLM-5.2 Locally

GLM-5.2 supports deployment with the following frameworks. Feel free to try them out:

- SGLang (v0.5.13.post1+) — see cookbook
- vLLM (v0.23.0+) — see recipes
- Transformers (v0.5.12+) — see transformers docs
- KTransformers (v0.5.12+) — see tutorial
- Unsloth (v0.1.47-beta+) — see guide
- For deployment on the `Ascend NPU` platform, inference frameworks such as vLLM-Ascend, xLLM and SGLang are supported — see here.

Footnote

- **Humanity's Last Exam (HLE) & other reasoning tasks**: We use sampling parameters of `temperature=1.0`, `top_p=0.95` for evaluation. We evaluate with a maximum generation length of `163,840` tokens. By default, we report the text-only subset; results marked with * are from the full set. For AIME, HMMT and IMOAnswerBench, we evaluate each question using the following system prompt: `Your response should be in the following format:\nExplanation: {your explanation for your final answer}\nExact Answer: {your succinct, final answer}\nConfidence: {your confidence score between 0% and 100% for your answer}.` We use GPT-5.5 (medium) as the judge model. For HLE-with-tools, we use a maximum context length of 300,000 tokens, with no context management strategy.

- **SWE-Bench Pro**: We run the SWE-Bench Pro suite with OpenHands using a tailored instruction prompt. Settings: `temperature=1`, `top_p=1`, `max_new_tokens=32k`, with a 400K context window.
- **NL2Repo**: We evaluated NL2Repo with `temperature=1.0`, `top_p=1.0`, and `max_new_tokens=48k` under 400k context. To prevent hacking, we use rule-based and a LLM-based judgement to prevent malicious behaviors (e.g., unauthorized pip or curl operations).
- **DeepSWE**: We run DeepSWE with the official pier evaluation framework and the mini-swe-agent harness (`temperature=1.0`, `top_p=1.0`, `timeout=2h`, 400K context). Each task is solved in an isolated container with 2 CPUs, 8 GB RAM, and no internet access.
- **ProgramBench**: We evaluate ProgramBench (200 instances) with Claude-Code 2.1.156 using `temperature=1.0, top_p=1.0, max_tokens=64000, max_turns=2000, sample_timeout=6h, reasoning_effort=max`, with a 400K context window. Each instance runs in a (4 CPUs, 8 GB RAM) sandbox with internet access disabled.
- **Terminal-Bench 2.1 (Terminus 2)**: We evaluate Terminal-Bench 2.1 with Terminus-2 framework using `parser=json`, `timeout=4h`, `temperature=1.0`, `top_p=1.0`, `max_new_tokens=48k`, `max_episodes=500`, with a 256K context window. Resource limits are capped at 4 CPUs and 8 GB RAM.
- **Terminal-Bench 2.1 (Claude Code)**: We evaluate in Claude Code 2.1.167 with `temperature=1.0, top_p=0.95, max_new_tokens=131072`. We override max_new_tokens to 128k via a transparent proxy, bypassing the 64k CLI cap to restore the configurability of `CLAUDE_CODE_MAX_OUTPUT_TOKENS`. We remove wall-clock time limits, while preserving per-task CPU and memory constraints. Scores are averaged over 5 runs.

- **MCP-Atlas**: All models were evaluated in think mode on the 500-task public subset with a 10-minute timeout per task. We use Gemini-3.0-Pro as the judge model for evaluation.
- **Tool-Decathlon**: We use the official evaluation service and set max_token to 128K.
- **FrontierSWE**: The evaluation was conducted by Proximal with 1M context length, max effort level, and 128K maximum output tokens. Dominance score reported as of 2026/06/16.
- **PostTrainBench**: The evaluation was conducted by PostTrainBench with 1M context length, max effort level, and 128K maximum output tokens.
- **SWE-Marathon**: The evaluation was conducted by Abundant AI with 1M context length, max effort level, and 128K maximum output tokens.

Citation

If you find GLM-5.2 useful in your research, please cite our technical report:

```
@misc{glm5team2026glm5vibecodingagentic,
      title={GLM-5: from Vibe Coding to Agen
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      url={https://arxiv.org/abs/2602.15763}
}
```